Supporting Statement

Bahar Barghbani

The University of Manchester

Resource-Efficient Execution of Data Centre-Grade Workloads with Lightweight Operating Systems on Emerging Capability Hardware

Supervisors: Dr Pierre Olivier and Dr Pavlos Petoumenos

I got interested in systems level research in my third semester in Sharif university. During my studies, I found a strong interest in understanding what is happening beneath the software level and how different components of a computer interact and cooperate. Moreover, my background in physics affects this interest in architecture level research, because I feel closer and more connected to it and I find it more physical. I am particularly interested in memory focues research and analyzing different mechanisms and innovations in the memory hierarchy to improve the performance of the processor. Furthermore, I also enjoy making use of software driven information to help with the functionality of the hardware. These reasons together are why I am interested in this particular project.

During Computer Sutrcture and Languages course in that semester, I became familiar with ISA and Assembely programming. I realized I enjoy understanding the operations beneath the software level. After that and with passing Computer Architecture and Operating Systems courses, I found myself very interested in analyzing the operations and interactions between different components of the computer. Therefore, I chose to deepen my understanding by conducting research in the area and I joined Professor Alberto Ros's research at the University of Murcia.

I started working with Prof. Ros since last winter, and I spent some time studying the literature and working on reproducing the results of an already published paper. However, I realized there are some discrepencies between the results and the ones reported in the published paper. After proposing several causes for this issue and collaborating with the PhD student who had worked on the project, we decided to move on and use an extended version of ChampSim, which was developed to include the Entangling Prefetcher. Currently, I am working on CVP workload traces to investiaget L2 cache placement for prefetched instructions. Right now, I am trying different policies such as different placement of block heads and block consecutive instructions, making use of the confidence of the entangled pair to decide on the pair's placement and working to add L2 prefetching to the overall mechanism and testing several settings to figure out a better way that works for these traces and outperforms a setting in which all prefetched instructions fill L1I cache.

More about my background, I am a bronze medalist in National Physics Olympiad. I believe, my background in physics was the root of my interest in low-level computer research, because I have a physical sense of what I am doing and what is happenning in the result. After starting my work with Professor Alberto and spending a six-week long on-site internship with Professor Ozcan Ozturk, which involved implementing compute-intensive and memory intensive benchmarks from scratch, as well as studying the state-of-the-art methods of using PMU for performance analysis [name of the papers here as well as changing this claim accordingly].

My main motivation to pursue a PhD is to dive deeper into working in memory focused research,

focusing on cache and front-end of the processor. In this project in project in particular, I am most excited about automating prefetching using LLM and using AI. Working in the intersection of Machine Learning and Memory Optimization is a field I can imagine myself working commitedly in.

Through my Computer Architecture course, I became familiar with memory and cache and learnt some of the basic concepts of memory hierarchy. After I was enrolled in the research in Entangling Prefetcher, I developed a strong interest in this area of research and improving processor performance through memory optimization.

The research with Dr. Ozturk is supposed to start more seriously and beyond preparation stages. The research question is whether we can come up with useful metrics dervied by engineering gathered data. The named data is supposed to be gathered from PMU registers.